After the Acquisition

Post-Merger Innovation Among Stayers in Pharma M&A

Tim Kühleis

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Paris School of Economics

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Pharma Mergers Raise Innovation Concerns That Remain Unanswered for Retained Inventors

Why this matters

- High concentration, frequent M&A, and exceptionally high R&D intensity make pharma the central industry for studying merger-innovation links
- FTC 2023 *Multilateral Pharmaceutical Merger Task Force*: "What is the full range of a pharma merger's effects on innovation?"

What the literature establishes

- Patenting falls 10–30% post-merger at the firm level (Ornaghi 2009; Haucap et al. 2019)
- Exit and departure rates spike; exiters and movers each lose ∼30% of expected career output (Cassi & Ornaghi 2026)
- **Open gap:** productivity, quality, and field trajectories of inventors who *stay* remain unexplored

Research Question

Do acquisitions reduce the patent quantity, citation-weighted quality, and technological focus of target-firm inventors who remain with the acquiring group — and does technological overlap between acquirer and target determine which channel dominates?

Two Channels Predict Opposing Effects — Post-Merger Frictions Dominate in Pharma

Post-merger frictions → lower output

- Integration frictions divert attention from discovery (Kapoor & Lim 2007)
- Loss of status reduces motivation for central inventors (Paruchuri et al. 2006)
- Disrupted knowledge networks limit access to tacit knowledge (Ernst & Vitt 2000)
- Weaker championing of early-stage projects (Hitt et al. 1991)

Internalized spillovers → higher output

- Mergers internalize informational externalities across R&D teams (Das et al. 2023)
- Hybrid target–acquirer coinvention raises citation impact (Li & Wang 2022)
- Bridge-building inventors gain ∼10% patenting boost (Sharoni 2026)

Net effect theoretically ambiguous. Frictions predicted to dominate in pharma: knowledge is tacit, integration costs are severe.

Hypotheses

H1 Acquisition reduces stayer patent *quantity*

H2 Acquisition reduces citation-weighted *quality*

H3 Stayers exhibit *TechDrift*: shift into acquirer's fields

H4 Negative H1/H2 effects attenuated when acquirer–target technological overlap is high (*DealSim*)

H5 Stayers who coinvent with acquirer-side inventors suffer smaller losses

Friction heterogeneity

Interaction	Sign
Age × After	+
Exclusivity × After	−
Big deal × After	−

One Million Inventor-Year Observations Spanning 513 Acquisitions Form the Empirical Foundation

Base data — Cassi & Ornaghi (2026, EER)

- EPO/PatStat (v. 2017) and Orbis/Zephyr (Bureau van Dijk)
- 311,358 inventors; 1,000,751 inventor $\times$ year obs., 1988–2015
- 513 pharma acquisitions; avg. 3.21 observations per inventor
- Upstream cleaning taken as given: disambiguation, original-applicant recovery, dynamic group construction

Own additions

- DuckDB/R pipeline: `inventor_year`, `inventor_ipc_year`, `group_ipc_year`
- OECD Patent Quality Indicators (PQII) linked via application ID: originality, radicalness, claims, composite score
- $TechDrift_{it}$ (H3): IPC cosine similarity to inventor's pre-merger baseline
- $DealSim_{c}$ (H4): pre-deal cosine similarity of acquirer and target IPC portfolios
- *Coinventor indicator* (H5): stayer files $\geq$1 post-deal joint patent with an acquirer-side inventor

TechDrift and DealSim both use binary IPC vectors. Declining TechDrift signals field misallocation; high DealSim signals technological overlap between merging firms. See formulas on final slide.

CS(2021) Tests Net Effects; DealSim and Coinventor Splits Isolate the Spillover Channel

Sample and estimator

- *Treated stayer:* $\geq$1 patent for target (5 pre-deal years), continues filing for acquiring group within $x \in \{3, 5\}$ years; acquirer-side inventors excluded
- *Control:* not-yet-treated — more comparable than structurally different never-treated firms
- Callaway & Sant'Anna (2021) group-time ATT, doubly robust

  Covariates: PrePatents, Age, Tenure, Exclusivity, FirmPatentStock

Outcomes tested (H1–H3)

- $Y_{it} \in \{$Patents, FracPatents, Citations, $\text{Quality}_{\text{PQII}}$, TechDrift$\}$
- Event study $\tau \in [-5, +5]$ validates parallel trends (Roth 2022)
- Aggregate ATT gives the main result for each outcome

Spillover-channel tests (H4–H5)

H4 — DealSim moderation:

- Re-estimate CS by DealSim tercile (low / mid / high)
- Prediction: |ATT| shrinks monotonically with overlap
- Robustness: TWFE with DealSim $\times$ After interaction

H5 — Coinventor split:

- Re-estimate CS for coinventing vs. non-coinventing stayers
- Prediction: non-coinventing stayers drive the negative ATT; coinventing stayers show attenuation
- Explains Cassi & Ornaghi's collaborator finding mechanically

CS(2021) estimating equations

Event study (dynamic ATT):

$$Y_{ict} = \sum_{\tau=-5}^{+5} \lambda_\tau \cdot \mathbf{1}[t - g_c = \tau] + \alpha_i + \xi_{ct} + \varepsilon_{ict}$$

Pre-treatment coefficients test parallel trends (Roth 2022).

Aggregate ATT:

$$Y_{ict} = \lambda \cdot \mathrm{After}_{ict} + \alpha_i + \xi_{ct} + \varepsilon_{ict}$$

α_i: inventor fixed effects ξ_{ct}: firm $\times$ year FE

$\lambda < 0$: post-acquisition decline in output.

Novel similarity measures

Field misallocation (H3):

$$\mathrm{TechDrift}_{it} = \frac{\mathbf{v}_i^t \cdot \mathbf{v}_i^0}{\left\|\mathbf{v}_i^t\right\| \left\|\mathbf{v}_i^0\right\|}$$

Declining values: inventor drifts into unfamiliar fields post-merger.

Technological overlap (H4):

$$\mathrm{DealSim}_c = \frac{\mathbf{v}_{\mathrm{target}}^{\mathrm{pre}} \cdot \mathbf{v}_{\mathrm{acq}}^{\mathrm{pre}}}{\left\|\mathbf{v}_{\mathrm{target}}^{\mathrm{pre}}\right\| \left\|\mathbf{v}_{\mathrm{acq}}^{\mathrm{pre}}\right\|}$$

High values: acquirer and target share technology fields — internalized spillovers more active.

Novel contributions

Quality and field outcomes

First quality-adjusted test of stayer performance using OECD PQII and TechDrift — prior work measures raw patent counts only.

Spillover-channel tests

H4 and H5 identify *when* and *through what conduit* internalized spillovers offset frictions, operationalizing Bourreau–Jullien–Lefouili at inventor level.

Identification

CS(2021) staggered DiD with doubly robust estimation, heterogeneity analysis, and Heckman selection correction at inventor level.

Policy implication: Headcount retention is not innovation retention.

With $180bn in patent revenues expiring 2027–28, pharma's acquisition-driven pipeline strategy carries an underappreciated cost: if acquisitions suppress stayer output — and the severity depends on deal design — current merger review frameworks cannot see it.